

UNLOCKING INDIAN CHIP'S SHAKTI

IIT Madras' Profs Patanjali and Gopalakrishnan Srinivasan tell us what it will take, from campus labs powering national ambitions

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At IIT Madras' Shakti lab, Professors Patanjali SLPSK and Gopalakrishnan Srinivasan aren't just participants in India's semiconductor journey, they are some of the most grounded narrators I've encountered while working on this cover story.

Their perspective strips away the rhetorical gloss around "chip sovereignty" and replaces it with something far more practical that we can all relate to. Stuff like talent pipelines, infrastructure constraints, licensing bottlenecks, and the slow, patient work of building a national hardware ecosystem from scratch.

If there's one theme that runs through during my interview conversation with both of them, it's this: India's semiconductor future will be decided less by grand announcements and more by whether the country can build deep, end-to-end capability – from architecture to application, and beyond.

Sovereignty isn't just about fabs

For Patanjali, who started as an intern

working in the Shakti processor lab and recently earned his PhD under IIT Madras' Director V Kamakoti's guidance, the idea of semiconductor sovereignty goes well beyond building fabs or chasing advanced nodes. It's about whether a country has the capability to design and deploy chips without relying on external gatekeepers.

"Sovereignty in computing essentially means that at no point we, as a country, are relying on a third party or an external entity for any aspect of our product development," Patanjali explains. "We want to make Shakti essentially a property of the entire country. So when we say sovereignty in computing, we essentially want to



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empower the country with the capabilities where there can be startups, and government agencies that can now think about building their own processor from scratch because the right people are already there in the market."

That definition reframes the debate. Sovereignty isn't just about fabs or policy incentives – it is about talent density and ecosystem maturity. It is about whether India can reach a point where building a processor becomes as routine as building software.

Gopalakrishnan or Gopal echoes this broader view. True sovereignty, he argues, exists only when every layer of the stack – design, IP, manu-